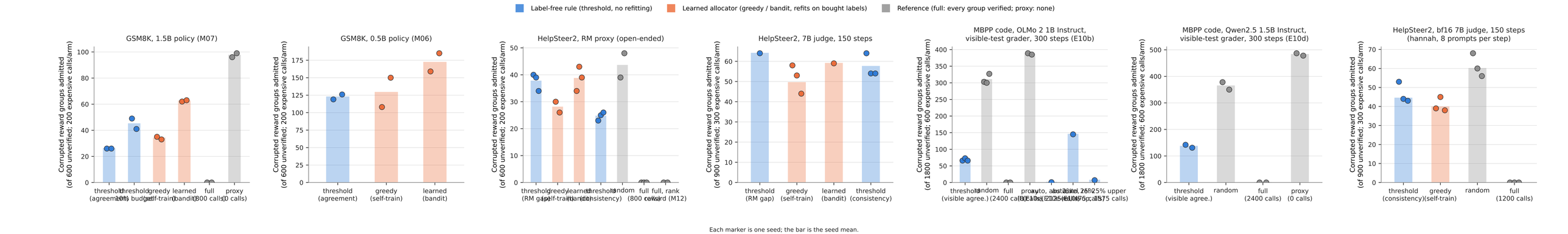

Label-free rule (threshold, no refitting)
Learned allocator (greedy / bandit, refits on bought labels)
Reference (full: every group verified; proxy: none)
GSM8K, 1.5B policy (M07)
Corrupted reward groups admitted (of 600 unverified; 200 expensive calls/arm)
0
20
40
60
80
100
threshold (agreement)
threshold (10% budget)
greedy (self-train)
learned (bandit)
full (800 calls)
proxy (0 calls)
GSM8K, 0.5B policy (M06)
Corrupted reward groups admitted (of 600 unverified; 200 expensive calls/arm)
0
25
50
75
100
125
150
175
threshold (agreement)
greedy (self-train)
learned (bandit)
HelpSteer2, RM proxy (open-ended)
Corrupted reward groups admitted (of 600 unverified; 200 expensive calls/arm)
0
10
20
30
40
50
threshold (RM gap)
greedy (self-train)
learned (bandit)
threshold (consistency)
random
full (800 calls)
full, rank reward (M12)
HelpSteer2, 7B judge, 150 steps
Corrupted reward groups admitted (of 900 unverified; 300 expensive calls/arm)
0
10
20
30
40
50
60
threshold (RM gap)
greedy (self-train)
learned (bandit)
threshold (consistency)
MBPP code, OLMo 2 1B Instruct, visible-test grader, 300 steps (E10b)
Corrupted reward groups admitted (of 1800 unverified; 600 expensive calls/arm)
0
50
100
150
200
250
300
350
400
threshold (visible agree.)
random
full (2400 calls)
proxy
25% upper
1575 calls)
MBPP code, Qwen2.5 1.5B Instruct, visible-test grader, 300 steps (E10d)
Corrupted reward groups admitted (of 1800 unverified; 600 expensive calls/arm)
0
100
200
300
400
500
threshold (visible agree.)
random
full (2400 calls)
proxy (0 calls)
HelpSteer2, bf16 7B judge, 150 steps (hannah, 8 prompts per step)
Corrupted reward groups admitted (of 900 unverified; 300 expensive calls/arm)
0
10
20
30
40
50
60
70
threshold (consistency)
greedy (self-train)
random
full (1200 calls)

Each marker is one seed; the bar is the seed mean.